

Rishiyesh KS
1925111466
KSR

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiment

Code of Conduct:

I Rishiyesh KS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.S. Rishiyesh

92/8

ASSESSMENT TOOL 1 — CO3

COMPUTER NETWORKS

Detail	Information
Course Name	Computer Networks
Course Code	CSA0720
Software	Cisco Packet Tracer, Wireshark

Student Name: Rishiykesh K S

Register No.: 192511466

QUESTION 1

Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

1. Given Data

Item	Configuration	Purpose / Evidence
PC1	192.168.1.10/24	Example host address
PC2	192.168.1.20/24	Example host address
Switch	2960	Layer-2 device

2. Problem Identification

Problem Identification: Two hosts are on the same local network, so communication depends on correct IP configuration and Layer-2 MAC addressing. The experiment must prove connectivity and then relate the observed Ethernet frame fields to switch-based forwarding.

3. Step-by-Step Procedure

Step 1: Place two PCs and one 2960 switch in Packet Tracer.

Step 2: Connect PC1 and PC2 to the switch using Copper Straight-Through cables.

Step 3: Open PC1 → Desktop → IP Configuration and set 192.168.1.10 with mask 255.255.255.0.

Step 4: Set PC2 to 192.168.1.20 with mask 255.255.255.0. A default gateway is not required for same-subnet communication.

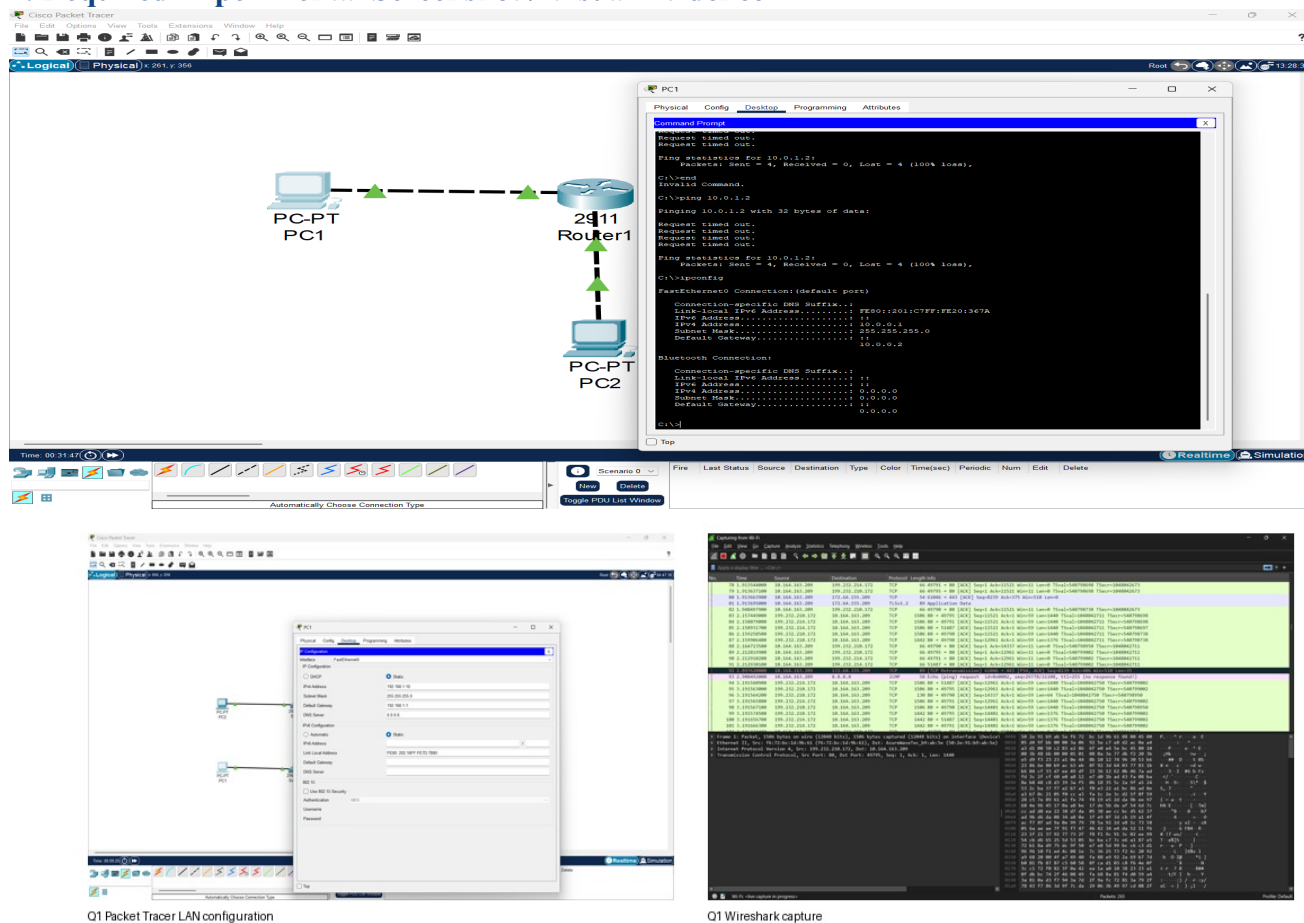
Step 5: From PC1 → Desktop → Command Prompt, run ping 192.168.1.20.

Step 6: Confirm successful Echo Replies and record the result.

Step 7: For Wireshark evidence, capture an equivalent real/virtual Ethernet exchange because Packet Tracer's internal simulation traffic is not directly captured by the host's Wireshark NIC capture.

Step 8: Use the Ethernet II section to identify Source MAC, Destination MAC and EtherType. Note that FCS may not be displayed in every host capture because NIC offloading can remove or hide it.

4. Required Experimental Screenshot / Visual Evidence



Q1 Packet Tracer LAN configuration

Q1 Wireshark capture

Actual Experimental Evidence — Question 1.

Evidence status: Actual experimental screenshot(s) inserted above.

5. Required Fields / Observations

Field / Packet	What to record or explain
Destination MAC	MAC address of the receiving interface
Source MAC	MAC address of the sending interface
EtherType	Identifies the carried network-layer protocol, e.g. IPv4
FCS	Ethernet error-detection field; may be absent in host captures

6. Analysis of Results

The successful ping proves that both hosts are correctly addressed and reachable in the same subnet. The switch uses the destination MAC address to select the outgoing port, while the Ethernet frame carries the Layer-3 packet.

7. Validation and Optimization

- Verify both hosts use the same subnet before interpreting the frame.
- Confirm ping success before selecting packets for analysis.
- Explain the relationship between IP addressing and MAC-based Layer-2 forwarding.

Justification: The answer combines configuration, packet-level evidence, interpretation, validation and optimization rather than only describing the protocol.

QUESTION 2

Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

1. Given Data

Item	Configuration	Purpose / Evidence
LAN 1	192.168.1.0/24	Gateway 192.168.1.1
Router G0/0	192.168.1.1/24	LAN 1 interface
Router G0/1	192.168.2.1/24	LAN 2 interface
LAN 2	192.168.2.0/24	Gateway 192.168.2.1

2. Problem Identification

Problem Identification: The two LANs are different IP networks. A switch alone cannot route between them, so the router must have a valid interface in each subnet and the hosts must use those interfaces as default gateways.

3. Step-by-Step Procedure

Step 1: Build PC1 → Switch0 → Router → Switch1 → PC2.

Step 2: Configure the router G0/0 as 192.168.1.1/24 and G0/1 as 192.168.2.1/24.

Step 3: On the router CLI use: enable → configure terminal → interface g0/0 → ip address 192.168.1.1 255.255.255.0 → no shutdown; repeat for g0/1 with 192.168.2.1.

Step 4: Configure PC1 as 192.168.1.10/24 with gateway 192.168.1.1.

Step 5: Configure PC2 as 192.168.2.10/24 with gateway 192.168.2.1.

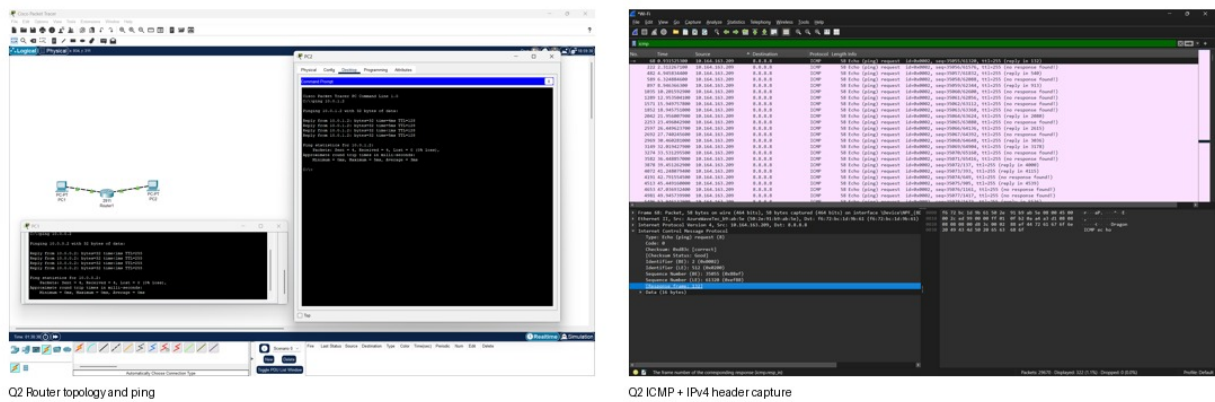
Step 6: Run show ip interface brief and verify the router interfaces are up.

Step 7: From PC1, run ping 192.168.2.10 and verify replies.

Step 8: Capture an equivalent ICMP exchange in Wireshark and inspect Internet Protocol Version 4.

Step 9: Record Source IP, Destination IP, TTL, Protocol and IHL/Header Length.

4. Required Experimental Screenshot / Visual Evidence



Actual Experimental Evidence — Question 2

Evidence status: Actual experimental screenshot(s) inserted above.

5. Required Fields / Observations

Field / Packet	What to record or explain
Source IP	Address of the originating host
Destination IP	Address of the remote host
TTL	Decreased at each router hop
Protocol	ICMP has protocol number 1
Header Length	IPv4 IHL $\times$ 4 bytes

6. Level-4 Analysis of Results

Successful end-to-end ping validates the addressing, default gateways and router forwarding. The IPv4 header identifies the logical source and destination, while TTL prevents indefinite packet circulation.

7. Validation and Optimization

- Check that both router interfaces are up/up.
- Verify each host uses the correct local router interface as its default gateway.
- Use destination IP and TTL to justify Network-layer forwarding.

Justification: The answer combines configuration, packet-level evidence, interpretation, validation and optimization rather than only describing the protocol.

QUESTION 3

Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

1. Given Data

Item	Configuration	Purpose / Evidence
Sender	192.168.1.10	Example source
Target	192.168.1.20	Example destination
ARP Request	Broadcast	Target MAC initially unknown
ARP Reply	Unicast	Returns target MAC

2. Problem Identification

Problem Identification: The sender knows the destination IP address but initially may not know the destination MAC address. ARP resolves this mapping so that the Ethernet frame can be addressed correctly on the local network.

3. Step-by-Step Procedure

Step 1: Connect multiple PCs to a switch.

Step 2: Assign PC1 = 192.168.1.10/24 and PC2 = 192.168.1.20/24.

Step 3: Clear the sender's ARP cache where supported, or restart the test so the first communication requires resolution.

Step 4: Start the packet capture for an equivalent real/virtual LAN exchange.

Step 5: Ping 192.168.1.20 from PC1.

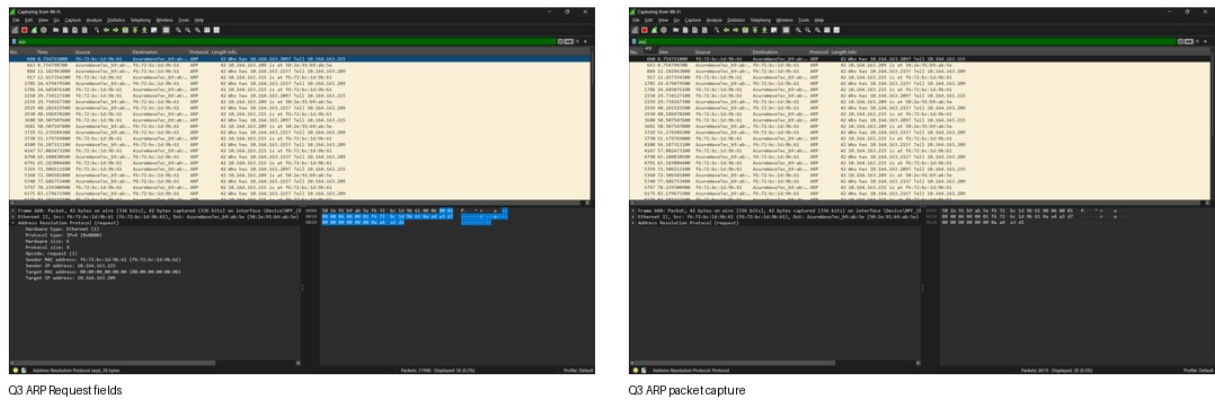
Step 6: Filter Wireshark with arp.

Step 7: Open the ARP Request and record sender IP/MAC and target IP/MAC fields.

Step 8: Open the ARP Reply and record the returned MAC address.

Step 9: Verify the resulting IP-to-MAC entry in the ARP table.

4. Required Experimental Screenshot / Visual Evidence



Actual Experimental Evidence — Question 3

Evidence status: Actual experimental screenshot(s) inserted above.

5. Required Fields / Observations

Field / Packet	What to record or explain
ARP Request	Broadcast query asking who owns the destination IP
Sender IP/MAC	Identifies the requesting host
Target IP	IP address whose MAC is required
ARP Reply	Provides the target IP-to-MAC mapping
ARP cache	Stores the mapping for later Ethernet delivery

6. Analysis of Results

The ARP Request discovers the destination MAC address. The Reply supplies the mapping, after which the sender can build an Ethernet frame addressed to the correct destination MAC.

7. Validation and Optimization

- Ensure the ARP cache is cleared or otherwise force a fresh resolution.
- Compare Request and Reply fields rather than only stating that ARP occurred.
- Verify that the learned mapping is available for subsequent Ethernet delivery.

Justification: The answer combines configuration, packet-level evidence, interpretation, validation and optimization rather than only describing the protocol.

QUESTION 4

Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

1. Given Data

Item	Configuration	Purpose / Evidence
PC-A	10.0.0.10/24	Ping source
Router	10.0.0.1 / 10.0.1.1	Two network interfaces
PC-B	10.0.1.10/24	Ping destination

2. Problem Identification

Problem Identification: The experiment must verify reachability through a router and distinguish an ICMP Echo Request from its Echo Reply. Timestamps are then used to calculate the round-trip delay.

3. Step-by-Step Procedure

Step 1: Build PC-A → Router → PC-B.

Step 2: Configure PC-A as 10.0.0.10/24 with gateway 10.0.0.1.

Step 3: Configure router interfaces as 10.0.0.1/24 and 10.0.1.1/24 and enable them.

Step 4: Configure PC-B as 10.0.1.10/24 with gateway 10.0.1.1.

Step 5: From PC-A, run ping 10.0.1.10 and confirm Echo Replies.

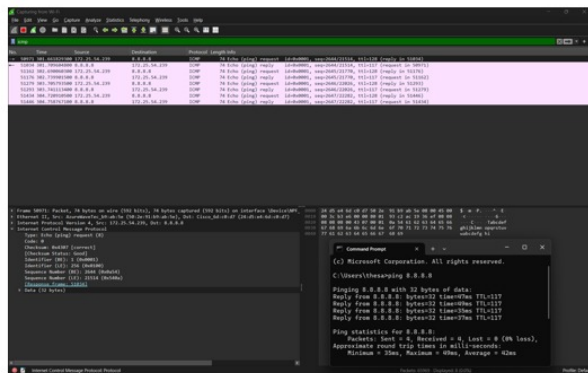
Step 6: Capture an equivalent ICMP exchange and filter with icmp.

Step 7: Identify Echo Request and Echo Reply and inspect Type, Code, Identifier and Sequence Number.

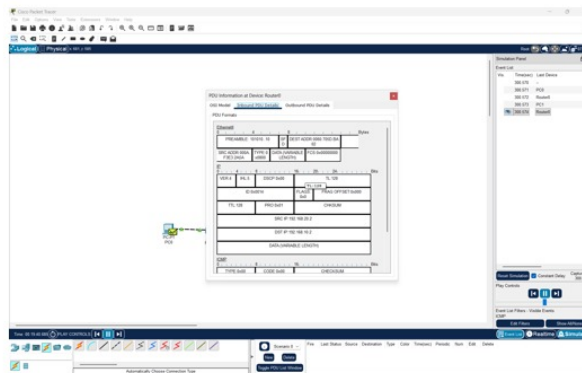
Step 8: Record the timestamps for one request and its matching reply.

Step 9: Calculate $RTT = \text{Reply timestamp} - \text{Request timestamp}$.

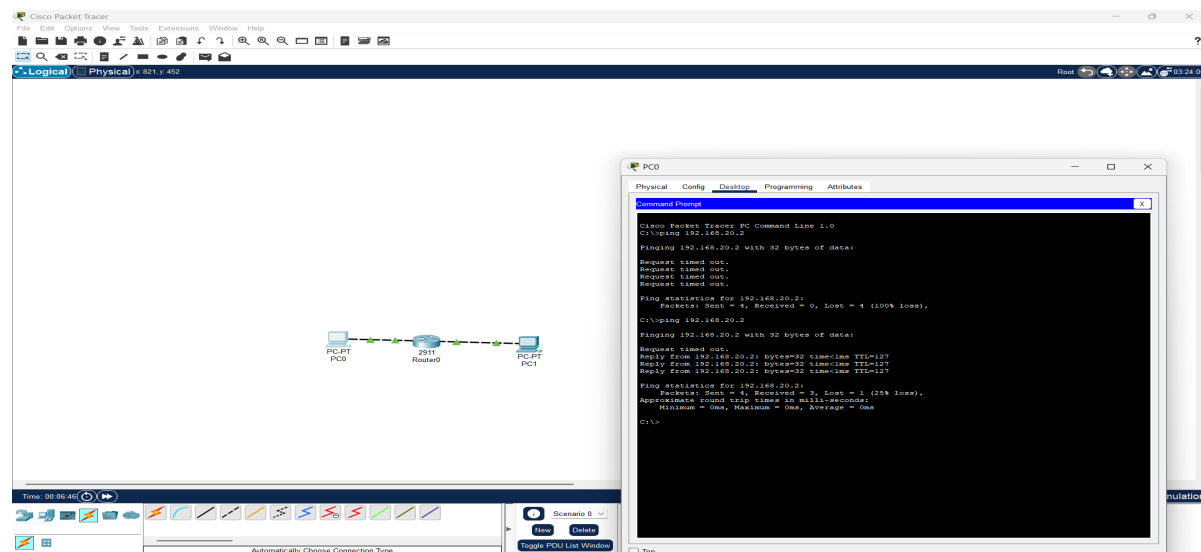
4. Required Experimental Screenshot / Visual Evidence



Q4 Final Wireshark ICMP evidence and RTT



Q4 Packet Tracer ICMP PDU



Actual Experimental Evidence — Question 4

Evidence status: Actual experimental screenshot(s) inserted above.

5. Required Fields / Observations

Field / Packet	What to record or explain
Echo Request	Type 8, normally Code 0
Echo Reply	Type 0, normally Code 0
Identifier	Associates the echo request/reply session
Sequence Number	Identifies individual echo messages
RTT	Reply time – Request time

6. Analysis of Results

ICMP provides a simple connectivity test. A successful Echo Reply proves that a return path exists, while RTT gives a measurable indication of response delay.

7. Validation and Optimization

- Match each Echo Reply with the correct request using Identifier and Sequence Number.
- Calculate RTT from the same capture timestamps.
- Repeat the test if necessary to observe RTT variation.

Justification: The answer combines configuration, packet-level evidence, interpretation, validation and optimization rather than only describing the protocol.

QUESTION 5

Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

1. Given Data

Item	Configuration	Purpose / Evidence
Client	192.168.1.10/24	Web browser
Server	192.168.1.100/24	HTTP service
TCP	Connection establishment	Three-way handshake
HTTP	Application layer	GET and response

2. Problem Identification

Problem Identification: The web client must first establish a reliable TCP connection before HTTP data is exchanged. The experiment therefore needs evidence of the SYN → SYN-ACK → ACK handshake followed by HTTP request and response.

3. Step-by-Step Procedure

Step 1: Place one client PC, one switch and one server.

Step 2: Configure the server as 192.168.1.100/24 and the client as 192.168.1.10/24.

Step 3: On Server → Services → HTTP, turn HTTP ON.

Step 4: From the client browser, enter http://192.168.1.100.

Step 5: Verify that the webpage loads successfully.

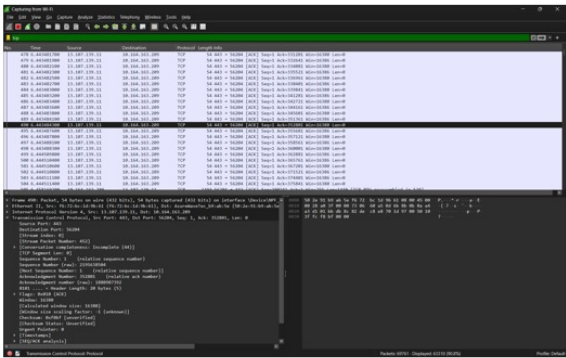
Step 6: Capture an equivalent HTTP/TCP exchange in Wireshark and filter with tcp.

Step 7: Identify SYN, SYN-ACK and ACK packets and record their sequence/acknowledgment behavior.

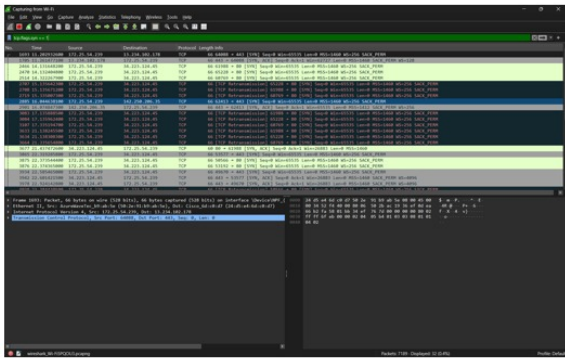
Step 8: Filter with http and identify the GET request and server response.

Step 9: Explain how TCP acknowledgments, sequence numbers, ordered delivery and retransmission support reliable HTTP communication.

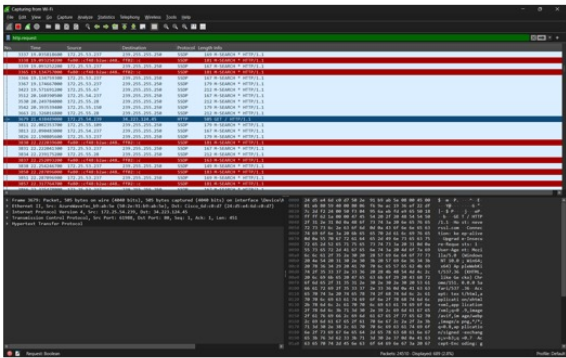
4. Required Experimental Screenshot / Visual Evidence



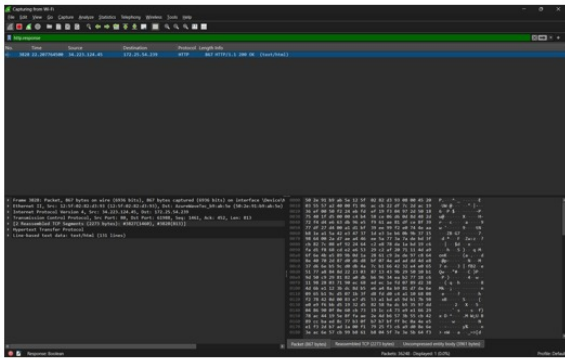
Q5 TCP capture / port 443



Q5 TCP SYN/SYN-ACK evidence



Q5 HTTP GET request



Q5 HTTP 200 OK response

Actual Experimental Evidence — Question 5.

Evidence status: Actual experimental screenshot(s) inserted above.

5. Required Fields / Observations

Field / Packet	What to record or explain
SYN	Client requests TCP connection
SYN-ACK	Server acknowledges and responds
ACK	Client completes connection establishment
HTTP GET	Client requests a web resource
HTTP Response	Server returns the requested resource

6. Analysis of Results

The TCP handshake establishes the connection before HTTP data transfer. TCP then provides sequencing, acknowledgments and retransmission mechanisms so the HTTP application can use a reliable ordered byte stream.

7. Validation and Optimization

- Confirm HTTP is enabled before troubleshooting the browser request.
- Separate TCP handshake packets from application-layer HTTP packets.
- Use sequence and acknowledgment behavior as evidence for reliable ordered delivery.

Justification: The answer combines configuration, packet-level evidence, interpretation, validation and optimization rather than only describing the protocol.